

Department of Computer Science & Engineering
Indian Institute of Technology Bombay
CS 787- Quiz I August 20, 2026 2-3:30 pm

1. First list three primary reasons for having a stub on the server side that we create automatically from an interface, and show how the solutions for these three aspects manifest in a pseudo c++-like code that is assumed to have been generated automatically from an interface. (10 Marks. You may draw a picture in support of your reasoning.)
2. Explain the entire solution for a push-push based event channel, including the interface and skeletal c++ code. (10 Marks. You may draw a picture in support of your reasoning.)
3. It is planned to develop an intermediate proxy for caching content in memory while accessing a file in read and write operations. Develop this theme giving relevant c++ code. Draw the picture of your class diagram design to help reading the code and to get clarity about your chosen design. (10 Marks)
4. Give one example c++ code to demonstrate how polymorphism works in multiple inheritance. Full code for the example is expected. (10 Marks)